

LATENT TREE PRIORS FOR BOND-FREE MOLECULAR FLOW MATCHING

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Development draft. Competitive advantage and submission readiness remain unestablished.

ABSTRACT

Molecular flow matching often starts from isotropic noise or a spatial prior constructed from a supplied chemical graph. We study spatial source distributions when atomic composition and electronic conditions are given but chemical bonds are unavailable. A latent dependence tree connects isotropic finite-length edge distributions, and a matrix-tree ratio gives the normalized marginal coordinate density after removing translation. Small composition-conditioned networks learn tree affinities directly from coordinates, without bond targets or gradients through sampled discrete trees. We integrate this source with a molecular flow and examine which source properties improve generation. In a bounded OMol25 development study, a fixed tree prior reduces fragmentation. Fresh draws from two matched training continuations also show higher graph acceptance than isotropic and covariance-matched single-Gaussian sources; comparison with a harmonic-tree mixture varies across continuations. An earlier weaker confirmation stream is retained. Learned affinities improve held-out source likelihood but have not established an increment over the fixed prior in final generation. A one-seed energy readout provides a node-network signal relative to an isotropic source. These results motivate separating source fit, spatial dependence and chemical utility; they do not establish equilibrium sampling or broad generator superiority.

1 INTRODUCTION

The starting distribution of a molecular generator specifies more than its noise scale: it determines which spatial dependencies the learned transport must create. Flow matching offers a flexible way to learn this transport (Lipman et al., 2023). Existing molecular implementations include joint generation of atom types, bonds and coordinates (Dunn & Koes, 2025), and conformer generation conditioned on a supplied molecular graph (Hassan et al., 2024). Here we consider a different, narrower task: generate coordinates conditional on a composition, total charge and spin multiplicity, without supplied bond labels.

A graph-informed harmonic source is useful when the graph is known. When bonds are unavailable, replacing them by one inferred graph commits to a discrete structure and can obscure the probability law of the coordinate source. Our construction instead averages over latent dependence trees. Every component has a normalized density on the translation-free coordinate space, and the full mixture admits tractable normalization and learning. The latent tree supplies a coordinate dependence pattern; it need not represent the final chemical bonds.

This construction combines established tree-ensemble identities with a finite-length, non-Gaussian spatial edge law. We do not claim a new matrix-tree theorem, tree-mixture principle, or physical work identity. The proposed method contribution is an explicit, learnable spatial source without supplied bonds and its integration into molecular flow matching. Whether this combination provides competitive practical value is the empirical question, rather than a consequence of normalization. Our present evidence supports a source-structure effect beyond the tested single-Gaussian controls, while an extra benefit of affinity learning remains unestablished.

2 RELATED WORK AND TASK BOUNDARY

Molecular flow matching. FlowMol3 studies joint molecular topology and geometry generation and reports improvements from self-conditioning, fake atoms and training geometry distortion (Dunn & Koes, 2025). We reuse its vector-field implementation with the repository’s electronic conditioning and coordinate-only research path. Published native FlowMol3 validity on its own benchmark is not directly comparable to our bond-free, charged/open-shell OMol25 development panel. ET-Flow uses a harmonic prior for conformer generation with a supplied molecular graph (Hassan et al., 2024). Our harmonic control is a controlled source ablation in the same backbone, not a reproduction of ET-Flow’s training or benchmark performance.

Tree distributions. Tractable factored distributions and learning over trees have a long history (Meila & Jaakkola, 2000). Spanning-tree likelihoods can characterize dependence without requiring the true population dependence graph to be a tree (Duan & Dunson, 2023). These are direct antecedents of our marginalization and affinity learning. Our application uses an intrinsic coordinate measure and normalized isotropic edge kernels, with no observed chemical graph.

Physical meaning. OMol25 provides molecular geometries with electronic conditions and calculated properties (Levine et al., 2025). Its empirical geometry distribution is not automatically a canonical ensemble for a specified temperature. We use an eSEN potential (Fu et al., 2025) for a separate output readout. Neither low potential energy nor a normalized source proves that the final generator follows a Boltzmann law.

3 A NORMALIZED LATENT-TREE SPATIAL SOURCE

Let $c = (z_1, \dots, z_N, Q, S)$ denote atomic numbers, integer total charge and spin multiplicity. We remove unweighted translation and use intrinsic Lebesgue measure on $\mathcal{H}_N = \{X \in \mathbb{R}^{N \times 3} : \sum_i x_i = 0\}$, of dimension $3(N - 1)$. For positive symmetric affinities $a_{ij}(c)$, define

$$p_\eta(T | c) = \frac{\prod_{(i,j) \in T} a_{ij}(c)}{\tau(a(c))}, \quad \tau(a) = \sum_{T \text{ spanning tree}} \prod_{(i,j) \in T} a_{ij}. \quad (1)$$

We sample T , draw independent edge displacements, integrate them on the tree, and subtract the coordinate mean. Weighted Wilson sampling supplies the tree law.

Finite-length edge law. Let ℓ_{ij} be the sum of the two covalent radii. Draw a uniform direction and a lognormal radius of mean ℓ_{ij} and log standard deviation s . The resulting normalized density against three-dimensional displacement measure is

$$h_{ij}(d) = \frac{\exp[-(\log r - \mu_{ij})^2 / (2s^2)]}{4\pi s \sqrt{2\pi} r^3}, \quad r = \|d\| > 0, \quad \mu_{ij} = \log \ell_{ij} - s^2/2. \quad (2)$$

The r^{-3} factor includes both the lognormal and solid-angle Jacobians. We fix $s = 0.2$ in these studies. This radial distribution is a source heuristic, not an electronic interaction potential. Exact coincidences have zero probability; the current differentiable density implementation rejects them as numerical inputs.

Proposition 1 (Intrinsic tree-mixture density). *If every h_{ij} is a normalized displacement density, then*

$$q_\eta(X | c) = N^{3/2} \frac{\tau(a_{ij}(c) h_{ij}(x_i - x_j))}{\tau(a(c))} \quad (3)$$

is normalized on $\mathcal{H}_N$. For an edge log affinity $\alpha_e = \log a_e$,

$$\frac{\partial \log q_\eta(X | c)}{\partial \alpha_e} = \Pr(e \in T | X, c) - \Pr(e \in T | c). \quad (4)$$

The proof is in Appendix A. Equation (3) is the marginal over all trees, rather than a density conditional on one sampled tree. Two grounded Laplacian determinants evaluate it in cubic time. Our implementation uses positive elimination in log space to avoid cancellation when edge weights span a large dynamic range.

Table 1: Graph-check passes out of 512 attempts per method. Confirmation changes continuation/data-order and sampling seeds but shares the warm checkpoint and development compositions. All attempts and validator exceptions are retained.

Training comparison	Warm	Gaussian	Fixed tree	Node tree	Pair tree
Unrestricted	78	76	94	92	93
Connected, first	66	105	131	121	111
Connected, confirmation	—	91	93	—	—

Learning affinities. The fixed reference has $a_{ij} = u_i u_j$, where $u_i = 0.05$ for H and the common terminal halogens and otherwise uses a simple default-valence-derived propensity. This is a heuristic and does not enforce chemical valence. A node network adds $b_i + b_j$ to $\log a_{ij}$; a pair network adds a symmetric pair score. Inputs are species embeddings, pooled composition, $\log N$, Q/N and $\log S$. No input coordinates or chemical bonds are provided to the affinity head at generation. Coordinate data fit its parameters through

$$\mathcal{L}_{\text{source}}(\eta) = \mathbb{E}_{(X,c) \sim \mathcal{D}_{\text{train}}} \left[-\frac{\log q_{\eta}(X | c)}{3(N-1)} \right]. \quad (5)$$

The node scores and pair score each use 2 tanh bounding. Consequently the summed node correction spans $[-4, 4]$ and the pair correction $[-2, 2]$; these are not strictly nested equal-range capacity ablations. Both learned priors retain the same fixed radial edge law.

4 INTEGRATION WITH FLOW MATCHING

After fitting the source, sample $X_0 \sim q_{\eta}(\cdot | c)$ and a training endpoint X_1 , apply the same type-preserving assignment/rotation coupling in all arms, and interpolate coordinates. Shared Haar augmentation preserves the rotation- and identical-label-permutation-invariant source law (Appendix B). The equivariant vector field learns the conditional path displacement. The pilot uses coordinate flow matching alone and no chemical-bond loss. Existing force/energy training interfaces remain available, but their independent-Gaussian velocity-to-score proxy is not valid for this correlated non-Gaussian pairing.

At generation, composition, charge and spin are fixed inputs. We integrate the displacement parameterization with 64 midpoint steps to $t = 1$ and add 0.025 Å centered Gaussian noise. This finite numerical sampler is the evaluated law. The continuous-flow density interface can use Equation (3) at an integrated preimage with its divergence correction. That fact does not provide the exact density of the evaluated midpoint-plus-noise sampler. We therefore make no absolute final-likelihood or importance-ESS claim.

5 DEVELOPMENT EXPERIMENTS

Matched continuations and target selection. All methods start from the same typed-matching warm checkpoint and receive 3,000 matched OMol25 training examples and AdamW updates at learning rate $2 \cdot 10^{-5}$. Node/pair priors fit 1,000 of those same examples; 128 separate train-split examples are used for source validation. No generated evaluation structure enters fitting. The maximum supported atom count remains 200; this development panel contains eight compositions with 9–20 atoms, including charged, open-shell and metal cases. Each method produces 64 outputs per composition.

The unrestricted pilot’s source-fitting sample contains 580 disconnected structures out of 1,000, whereas the desired output criterion is a single connected structure. We preserve that pilot and apply a shared training-only connected/nonoverlapping selection to the next comparison. The first continuation selects 3,128 examples after examining 7,130; the confirmation examines 7,019 with a new data-order seed. Contact and overlap cutoffs are fixed at 1.25 and 0.6 times the sum of covalent radii. This changes the target domain and is not evidence of an improvement on all unrestricted OMol25 geometries.

Table 2: Fresh moment-control outputs: graph check passed /512. All methods share training data and update budgets within a continuation.

	Continuation	Unit Gaussian	Covariance Gaussian	Harmonic tree	Shell tree
1		103	106	112	136
2		94	96	128	118

Source fit and output utility differ. Connected-target held source NLL per degree of freedom is 1.59362 for fixed, 1.58443 for node and 1.57507 for pair. Table 1 shows that this ordering does not translate into higher graph acceptance. Fixed minus Gaussian graph pass rate is +5.08 percentage points with a conditional paired 95% interval [1.17, 8.79] initially, but +0.39 with interval [-3.52, 4.10] in confirmation. This original confirmation stream alone did not establish a graph advantage; we retain it alongside the fresh control-study draws reported below.

Fragmentation shows a more consistent mechanism: disconnected counts fall from 192 to 162 in the first continuation and 243 to 194 in confirmation. Geometry-supported counts rise from 314 to 340 and 266 to 312. These checks neither certify the requested electronic state nor enforce valid chemical valence. The legacy RDKit charged-fragment graph perception has radical and metal limitations; exceptions are reported without retroactively revising any graph count or decision.

Energy readout. We evaluate every output of all five first-connected-run methods with the same frozen eSEN checkpoint at its original charge and spin. Each structure and its inversion are scored, totaling 5,120 raw queries. Define $E_+(X) = [E(X) + E(-X)]/2$. Mean differences are taken within compositions, then averaged with equal composition weight. Node minus Gaussian is -0.624 eV with conditional interval [-1.084, -0.144], but node minus fixed is -0.229 eV with interval [-0.739, 0.269]. Fixed minus Gaussian is -0.395 eV with interval [-0.907, 0.135]. This is an unreplicated node-network signal relative to Gaussian, not an established learned increment over the fixed prior. The Re-containing case worsens for fixed and pair and is retained. No output was chosen by energy.

Second-moment controls. Two additional controls isolate source properties. A harmonic-tree mixture keeps the exact tree law and uses Gaussian edge covariance $v_{ij}I = \ell_{ij}^2 \exp(s^2)I/3$. It matches the shell-tree covariance for every tree. A single Gaussian uses a fixed 128-tree average of analytic covariances, symmetrized over identical species. Against an independent 8,192-tree calibration, the covariance approximation differs by 2.4–6.1% in Frobenius norm and at most 3.5% in trace on the eight compositions. The complete experiment includes 4,096 fresh outputs from four source variants on each of two continuations. The Gaussian and shell references reuse the same two checkpoints as the earlier connected studies, with new generation streams; the two new source controls receive matching training examples and budgets. These are not two additional shell-model fits. Table 2 reports the complete fresh comparison. The shell minus covariance Gaussian graph pass-rate differences are +5.86 percentage points [2.15, 9.38] and +4.30 percentage points [0.59, 8.01], respectively. The shell minus harmonic tree graph pass-rate differences are +4.69 percentage points [0.98, 8.40] and -1.95 percentage points [-5.66, 1.76], respectively. These intervals condition on fitted models and the same development compositions. The controls do not test an additional learned-affinity increment.

Both new shell-versus-single-Gaussian comparisons have positive conditional intervals. Against the harmonic mixture the sign changes across continuations. This supports a useful source structure beyond the tested Gaussian covariance control, but does not establish a repeatable benefit of the narrow radial law. The covariance control’s finite calibration error remains a limitation of exact moment attribution. No new physical queries are made in this experiment.

Uncertainty and remaining evidence. The intervals resample paired draws within each of the eight fixed compositions, conditional on the fitted models, without multiplicity adjustment. They do not quantify new-composition or pretraining-seed uncertainty. Broad held-composition evaluation, comparisons with independently trained learned generators and a replicated useful learned-source increment remain outstanding. A favorable source likelihood or one conditional energy interval cannot replace those experiments.

6 DISCUSSION

The current framework makes a bond-free source probability law explicit and learnable. The new controls support a useful source structure beyond a single Gaussian approximation, while affinity learning’s downstream value remains unresolved. Narrow finite-length edges are not consistently better than a harmonic tree mixture. One concrete open question is whether the flow should retain the sampled tree as conditioning, instead of receiving its coordinates alone. That architecture has not been tested here. The present results do not establish a calibrated energy distribution or a completed competitive benchmark.

REPRODUCIBILITY STATEMENT

Frozen protocols record data-order and generation seeds, source checkpoint hashes, all attempted structures, electronic conditions and oracle costs. Current audits replay 10,240 source draws and structural readouts and 5,120 stored physical rows; they do not independently rerun optimizers, final neural generation or the oracle. The 4,096-output moment-control audit is complete. Implementation tests cover normalization, Jacobians, weighted-tree sampling, symmetry, density gradients and source-law guards. The anonymous release package and final author verification remain outstanding.

AI USE STATEMENT

AI tools assisted research exploration, literature checks, mathematical derivation, implementation, testing, experiment orchestration, analysis and drafting. This is an internal development version; final human-author review remains outstanding.

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A NORMALIZATION AND AFFINITY-GRADIENT PROOF

For a tree T , let $B_T \in \mathbb{R}^{(N-1) \times N}$ be an oriented incidence matrix and $U \in \mathbb{R}^{N \times (N-1)}$ an orthonormal basis of the centered subspace. Because $B_T \mathbf{1} = 0$, $(B_T U)(B_T U)^\top = B_T B_T^\top$. Cauchy–Binet expresses $\det(B_T B_T^\top)$ as the sum of squares of its N maximal minors. Each minor of a tree incidence matrix is ± 1 , so the determinant is N . The absolute Jacobian in one spatial dimension is therefore $\sqrt{N}$, and in three dimensions it is $N^{3/2}$. Consequently the component density is $q(X \mid T, c) = N^{3/2} \prod_{e \in T} h_e(B_T X)$ and integrates to one. Averaging with Equation (1) yields Equation (3) and preserves normalization. Differentiating the log of each finite positive tree partition with respect to α_e gives its edge-inclusion probability, proving Equation (4). These are applications of standard incidence and tree-ensemble identities.

B SOURCE INVARIANCE UNDER SYMMETRY COUPLING

Let G be the compact group of proper rotations and permutations of identical conditioning labels. Suppose q_0 is G -invariant, $X \sim q_0$, and a pairing algorithm selects $g(X, Y) \in G$ using a target Y . If H is an independent Haar draw, then conditional on the orbit of X and on the selected g , HgX has the uniform orbit distribution of HX , by Haar invariance. Integrating over the unchanged source-orbit distribution recovers q_0 . This argument permits a non-Gaussian source and explains the shared augmentation used in the pairing. It does not imply independence of source and target after pairing or justify the independent-Gaussian velocity-to-score formula.

C COVARIANCE CONTROLS

For $R = \ell \exp(sZ - s^2/2)$ with $Z \sim N(0, 1)$, $\mathbb{E}R^2 = \ell^2 \exp(s^2)$. A uniform direction has second moment $I/3$, so each edge has covariance $vI = \ell^2 \exp(s^2)I/3$. Let A_T map edge coordinates to centered node coordinates. Independence gives $C_T = A_T \text{diag}(v_e) A_T^\top$. Replacing the edge vectors by independent Gaussians of covariance $v_e I$ leaves C_T unchanged and, since all component means are zero, leaves $\mathbb{E}_T C_T$ unchanged. The single-Gaussian control uses a finite, composition-seeded average of these matrices. Averaging this matrix over within-species permutations preserves positive definiteness on the centered subspace and yields species invariance. The Gaussian is exactly normalized for its chosen covariance; equality to the infinite tree-mixture covariance remains approximate.